

FCAI ANU Materials – Semi Seniors – Computer Graphics – Lecture 1

Introduction to Computer Graphics

Batch 1 – Questions 1 to 20

Q1. What are the two main objectives stated at the start of the lecture?

- a) Learn a graphics API and write shader code
- b) Understand broadly what computer graphics is about, and implement the first algorithm for making images of 3D shapes
- c) Study hardware architecture and GPU pipelines
- d) Review linear algebra and calculus

Answer: b) Understand broadly what computer graphics is about, and implement the first algorithm for making images of 3D shapes

Q2. According to the lecture's definition, computer graphics is:

- a) The use of computers to synthesize visual information
- b) The study of pixel manipulation only
- c) A branch of computer hardware engineering
- d) The science of image compression

Answer: a) The use of computers to synthesize visual information

Q3. The lecture later revises the definition of graphics to include:

- a) Only visual information
- b) The use of computation to turn digital information into sensory stimuli
- c) Only sound and touch
- d) Only 3D geometry

Answer: b) The use of computation to turn digital information into sensory stimuli

Q4. Roughly what fraction of the brain is dedicated to visual processing, according to the lecture?

- a) About 5%
- b) About 30%
- c) About 60%
- d) About 90%

Answer: b) About 30%

Q5. Why are eyes described as important in the lecture?

- a) They are the slowest input to the brain
- b) They are the highest-bandwidth port into the head
- c) They only process color, not shape
- d) They are not relevant to computer graphics

Answer: b) They are the highest-bandwidth port into the head

Q6. Which historical device is mentioned as an early computer (1945)?

- a) Sketchpad
- b) ENIAC
- c) Apple II
- d) IBM PC

Answer: b) ENIAC

Q7. Punch cards, as referenced in the lecture, stored roughly how much data?

- a) ~12 bytes
- b) ~120 bytes
- c) ~1.2 KB
- d) ~12 KB

Answer: b) ~120 bytes

Q8. Sketchpad, an early interactive graphics system, was created by:

- a) Alan Turing
- b) Ivan Sutherland
- c) John von Neumann
- d) Douglas Engelbart

Answer: b) Ivan Sutherland

Q9. In what year was Sketchpad created?

- a) 1945
- b) 1955
- c) 1963
- d) 1980

Answer: c) 1963

Q10. An 8K monitor (7680x4320) produces an image of roughly what size, per the lecture?

- a) ~9.5 MB
- b) ~95 MB
- c) ~950 MB
- d) ~9.5 GB

Answer: b) ~95 MB

Q11. According to the lecture, a 2020 VR headset with two 2160x2160 displays at 90Hz produces data at roughly:

- a) 2.3 MB/s
- b) 23 MB/s
- c) 2.3 GB/s
- d) 23 GB/s

Answer: c) 2.3 GB/s

Q12. Which of the following is NOT listed as an application area of computer graphics in the lecture?

- a) Entertainment (movies, games)
- b) Architecture
- c) Scientific/mathematical visualization
- d) Database indexing

Answer: d) Database indexing

Q13. Which visualization type is specifically mentioned alongside scientific visualization?

- a) Medical/anatomical visualization
- b) Financial visualization
- c) Network visualization
- d) Weather visualization

Answer: a) Medical/anatomical visualization

Q14. Which of the following is listed under the 'Theory' foundations of computer graphics?

- a) Sampling & aliasing
- b) Compiler design
- c) Operating systems
- d) Database theory

Answer: a) Sampling & aliasing

Q15. Which of the following is listed under the 'Theory' foundations of computer graphics?

- a) Radiometry & light transport
- b) Network protocols
- c) File systems
- d) Cryptography

Answer: a) Radiometry & light transport

Q16. Which of the following is listed under the 'Systems' foundations of computer graphics?

- a) Parallel, heterogeneous processing
- b) Radiometry
- c) Perception
- d) Sampling & aliasing

Answer: a) Parallel, heterogeneous processing

Q17. In the cube-modeling activity, the cube is assumed to be centered at:

- a) (1,1,1)
- b) The origin (0,0,0)
- c) (2,2,2)
- d) (-1,-1,-1)

Answer: b) The origin (0,0,0)

Q18. What are the dimensions of the cube used in the modeling activity?

- a) 1x1x1
- b) 2x2x2
- c) 3x3x3
- d) 4x4x4

Answer: b) 2x2x2

Q19. How many vertices does the cube in the activity have?

- a) 4
- b) 6
- c) 8
- d) 12

Answer: c) 8

Q20. How many edges does the cube in the activity have?

- a) 6
- b) 8
- c) 10
- d) 12

Answer: d) 12

Batch 2 – Questions 21 to 40

Q21. In the cube-drawing activity, what is the basic two-step strategy for turning a 3D cube into a 2D image?

- a) Rotate the cube, then scale it
- b) Map 3D vertices to 2D points, then connect them with straight lines
- c) Apply color, then shade the faces
- d) Compute normals, then apply lighting

Answer: b) Map 3D vertices to 2D points, then connect them with straight lines

Q22. The lecture explains perspective projection using which simple camera model?

- a) Lens camera
- b) Pinhole camera
- c) Fisheye camera
- d) Orthographic camera

Answer: b) Pinhole camera

Q23. In perspective projection, objects appear smaller as they:

- a) Get closer to the camera
- b) Get further away from the camera
- c) Rotate faster
- d) Change color

Answer: b) Get further away from the camera

Q24. In the side-view derivation, the image point is called:

- a) $p = (x, y, z)$
- b) $q = (u, v)$
- c) $c = (2, 3, 5)$
- d) $s = \text{slope}$

Answer: b) $q = (u, v)$

Q25. Under the assumption that the camera has unit size with origin at the pinhole c , the vertical image coordinate v is derived as:

- a) $v = y + z$
- b) $v = y/z$ (the slope y/z)
- c) $v = y * z$
- d) $v = z/y$

Answer: b) $v = y/z$ (the slope y/z)

Q26. Following the same logic, the horizontal image coordinate u is given by:

- a) $u = x/z$
- b) $u = z/x$
- c) $u = x * z$
- d) $u = x + z$

Answer: a) $u = x/z$

Q27. In the 'draw the cube' activity, what camera position is assumed?

- a) $c = (0, 0, 0)$
- b) $c = (1, 1, 1)$
- c) $c = (2, 3, 5)$
- d) $c = (5, 3, 2)$

Answer: c) $c = (2, 3, 5)$

Q28. To project each 3D vertex to 2D in the activity, the first step is to:

- a) Divide (x,y) by z directly
- b) Subtract the camera position c from the vertex to get (x,y,z)
- c) Multiply the vertex by the camera position
- d) Add the camera position to the vertex

Answer: b) Subtract the camera position c from the vertex to get (x,y,z)

Q29. After subtracting the camera position, the second step to get (u,v) is to:

- a) Multiply (x,y) by z
- b) Divide (x,y) by z
- c) Add z to (x,y)
- d) Take the square root of z

Answer: b) Divide (x,y) by z

Q30. Once the two endpoints of an edge have been projected to (u_1,v_1) and (u_2,v_2) , what does the algorithm do next?

- a) Fill the polygon between them
- b) Draw a line between the two 2D points
- c) Compute a normal vector
- d) Apply a color gradient

Answer: b) Draw a line between the two 2D points

Q31. According to the lecture, what did the cube-drawing exercise successfully demonstrate?

- a) Turning visual information into digital information
- b) Turning purely digital information into purely visual information using a completely algorithmic procedure
- c) Compressing an image
- d) Simulating lighting

Answer: b) Turning purely digital information into purely visual information using a completely algorithmic procedure

Q32. The lecture describes a raster display using which common abstraction?

- a) A continuous vector canvas
- b) A 2D grid of pixels, each with a color value
- c) A single scanning laser beam
- d) A 1D array of intensities

Answer: b) A 2D grid of pixels, each with a color value

Q33. The process of converting a continuous object (like a line) into a discrete pixel-grid representation is called:

- a) Aliasing
- b) Rasterization
- c) Tessellation
- d) Quantization

Answer: b) Rasterization

Q34. Which rule for choosing which pixels to light up for a line is specifically named in the lecture, and used by modern GPUs?

- a) The midpoint rule
- b) The diamond rule

- c) The Bresenham rule
- d) The scanline rule

Answer: b) The diamond rule

Q35. Under the diamond rule, a pixel is lit up if:

- a) The line passes through the pixel's associated diamond
- b) The line touches any corner of the pixel
- c) The pixel is closer to the camera
- d) The pixel's color matches the line's color

Answer: a) The line passes through the pixel's associated diamond

Q36. Why is naively checking every pixel in the image to rasterize a line considered inefficient?

- a) It only works for horizontal lines
- b) It costs $O(n^2)$ work in the number of image pixels, versus at most $O(n)$ pixels actually lit up
- c) It cannot represent color
- d) It requires floating-point hardware that doesn't exist

Answer: b) It costs $O(n^2)$ work in the number of image pixels, versus at most $O(n)$ pixels actually lit up

Q37. In the incremental line rasterization algorithm described, the slope s of the line is computed as:

- a) $s = (u_2 - u_1) / (v_2 - v_1)$
- b) $s = (v_2 - v_1) / (u_2 - u_1)$
- c) $s = (u_2 + u_1) / (v_2 + v_1)$
- d) $s = u_2 * v_2$

Answer: b) $s = (v_2 - v_1) / (u_2 - u_1)$

Q38. The easy special case handled by the incremental algorithm assumes:

- a) $u_1 > u_2$ and $v_1 > v_2$
- b) $u_1 < u_2$, $v_1 < v_2$, and $0 < s < 1$
- c) The line is vertical
- d) The slope is greater than 1

Answer: b) $u_1 < u_2$, $v_1 < v_2$, and $0 < s < 1$

Q39. In the incremental algorithm's loop, what is updated on every iteration as u increases by 1?

- a) v is incremented by s , then rounded to draw the pixel
- b) u is divided by s
- c) The camera position is updated
- d) The color is incremented

Answer: a) v is incremented by s , then rounded to draw the pixel

Q40. The lecture notes that although the incremental algorithm is easy to implement, it is:

- a) Exactly how lines are drawn in modern software/hardware
- b) Not how lines are actually drawn in modern software/hardware
- c) Only usable for 3D lines
- d) The fastest possible method

Answer: b) Not how lines are actually drawn in modern software/hardware

Batch 3 – Questions 41 to 48

Q41. After completing the simple line-drawing algorithm for the cube, what does the lecture say is needed for more realistic pictures?

- a) Only better monitors
- b) A richer model of the world, including geometry, materials, lights, cameras, and motion
- c) Faster punch card readers
- d) More pixels per inch only

Answer: b) A richer model of the world, including geometry, materials, lights, cameras, and motion

Q42. Which of the following is explicitly listed as part of the 'richer model of the world' needed for realism?

- a) Materials
- b) Punch cards
- c) Compilers
- d) Databases

Answer: a) Materials

Q43. Which of the following is explicitly listed as part of the 'richer model of the world' needed for realism?

- a) Motion
- b) Networking
- c) Cryptography
- d) Operating systems

Answer: a) Motion

Q44. The lecture frames the whole cube-drawing exercise as fundamentally illustrating:

- a) What computer graphics is all about
- b) How to compress an image
- c) How compilers work
- d) How networks transmit images

Answer: a) What computer graphics is all about

Q45. What footnote does the lecture add about the idea of a pixel as 'a little square'?

- a) It is completely accurate and never questioned
- b) The notion will be strongly challenged later in the course
- c) Pixels are always circular
- d) Pixels do not have color

Answer: b) The notion will be strongly challenged later in the course

Q46. The lecture references a SIGGRAPH trailer to make which point?

- a) That computer graphics is a narrow, niche field
- b) That even the broadened definition of graphics is still too narrow
- c) That SIGGRAPH only covers hardware
- d) That graphics has not changed since the 1960s

Answer: b) That even the broadened definition of graphics is still too narrow

Q47. The lecture mentions turning digital information into physical matter as an example of graphics evolving beyond:

- a) Just turning on pixels
- b) Just writing text

- c) Just playing sound
- d) Just 2D drawing

Answer: a) Just turning on pixels

Q48. Where does the lecture say students can find all logistics for the course?

- a) In the lecture slides only
- b) On the course webpage
- c) By email only
- d) In the textbook appendix

Answer: b) On the course webpage

Wish you all the best!
Fathy.